Identity of precategories is isomorphism of precategories, by univalence

The Univalent Foundations Program

hott-book:categories.tex:local-snapshot-20261011; CC BY-SA 3.0 Unported

Editorial note. The counted statement is the author's 'ct:cat-eq-iso' (source lines 830-834, Lemma 9.4.15); the proof environment starts on the next line, so the pairing is by adjacency.

Editorial note. No context span is retained: the statement and proof use no label from outside the retained range.

Editorial note. The retained counter values (chapter 9, section 4, theorem offset 14) make the counted lemma print as Lemma 9.4.15, matching the original book numbering taken from the label-number dump of the pinned snapshot.

Editorial note. The retained preamble is the chapter's own retained starter preamble for this pinned source.

Editorial scope. One original result of the HoTT Book, counted once: identity of precategories is isomorphism of precategories (Lemma 9.4.15).

Allowed standard prerequisites (editorial).

- Structure identity for precategories. A precategory is a nested sigma type of object types, hom-types and structure, so an identity between precategories unfolds into componentwise identities together with coherence conditions.. It is the unfolding performed at source lines 836-843.
- Univalence. An equivalence of types gives a path between them, and transport along that path computes by applying the equivalence.. It converts the componentwise identities into equivalences at source line 845.
- Coherence by mere propositions. The identity types of hom-sets are mere propositions, so the coherence conditions need no further data.. It is the remark at source line 844.

Lemma 9.4.15. *If A and B are precategories, then the function*

$$(A = B) \to (A \cong B)$$

(defined by induction from the identity functor) is an equivalence of types.

Proof. As usual for dependent sum types, to give an element of $A = B$ is equivalent to giving

- an identity $P_0 : A_0 = B_0$,
- for each $a, b : A_0$, an identity

$$P_{a,b} : \hom_A(a, b) = \hom_B(P_{0*}(a), P_{0*}(b)),$$

- identities $(P_{a,a})_*(1_a) = 1_{P_{0*}(a)}$ and $(P_{a,c})_*(gf) = (P_{b,c})_*(g) \circ (P_{a,b})_*(f)$.

(Again, we use the fact that the identity types of hom-sets are mere propositions.) However, by univalence, this is equivalent to giving

- an equivalence of types $F_0 : A_0 \simeq B_0$,
- for each $a, b : A_0$, an equivalence of types

$$F_{a,b} : \hom_A(a, b) \simeq \hom_B(F_0(a), F_0(b)),$$

- and identities $F_{a,a}(1_a) = 1_{F_0(a)}$ and $F_{a,c}(gf) = F_{b,c}(g) \circ F_{a,b}(f)$.

But this consists exactly of a functor $F : A \to B$ that is an isomorphism of categories. And by induction on identity, this equivalence $(A = B) \simeq (A \cong B)$ is equal to the one obtained by induction. □